Raw tBLASTn evidence underlying assembly-level gene recovery

Trypanosoma equiperdum IVM-t1 mitochondrial-maintenance gene panel

HSP	Query protein	IVM-t1 contig / accession	Alignment length (aa)	Query coverage (%)	Identity (%)	Mismatches	Gap opens	Query coordinates	IVM-t1 coordinates	E-value	Bit score
ATPF1A											
1	Tb927.7.7420:mRNA-p1	CM010682.1	404	77.8	**25.99**	270	13	76–465	2170537–2171703	1.67×10^{-20}	96.3
2	Tb927.7.7420:mRNA-p1	CM010674.1	330	63.6	**24.55**	218	8	119–437	270396–271325	4.81×10^{-13}	72.4
3	Tb927.7.7420:mRNA-p1	CM010675.1	255	49.1	**27.45**	153	6	173–410	221987–221268	1.37×10^{-12}	70.9
ATPF1B											
1	Tb927.3.1380:mRNA-p1	CM010674.1	519	**100.0**	**100.00**	0	0	1–519	270096–271652	0.00	969.0
2	Tb927.3.1380:mRNA-p1	CM010675.1	320	61.7	29.38	204	9	140–442	222014–221070	1.18×10^{-24}	108.0
3	Tb927.3.1380:mRNA-p1	CM010682.1	360	69.4	27.78	242	5	98–444	2170657–2171721	1.47×10^{-18}	89.4
ATPF1G											
1	Tb927.10.180:mRNA-p1	CM010681.1	305	**100.0**	**99.34**	2	0	1–305	52732–53646	0.00	625.0
ATOM40											
1	Tb927.9.9660:mRNA-p1	CM010680.1	351	**100.0**	**100.00**	0	0	1–351	963059–962007	0.00	700.0
POLIB											
1	Tb927.11.4690:mRNA-p1	QSBY01000015.1	1,404	**100.0**	**99.93**	1	0	1–1404	825327–821116	0.00	2,882.0
2	Tb927.11.4690:mRNA-p1	CM010682.1	373	26.6	37.27	198	9	1028–1367	900286–901395	1.23×10^{-59}	227.0
3	Tb927.11.4690:mRNA-p1	CM010682.1	473	33.7	27.27	249	13	307–692	897535–898929	1.48×10^{-41}	168.0
4	Tb927.11.4690:mRNA-p1	CM010678.1	346	24.6	35.84	192	8	1037–1358	1206397–1207416	5.37×10^{-52}	202.0
TAC102											
1	Tb927.7.2390:mRNA-p1	CM010678.1	648	**68.1**	**99.38**	4	0	1–648	757205–755262	0.00	1,300.0
2	Tb927.7.2390:mRNA-p1	CM010678.1	219	**23.0**	**99.54**	1	0	733–951	755009–754353	6.48×10^{-107}	365.0

HSP, high-scoring segment pair. Query coverage is calculated as `alignment length / full-length query protein × 100` for each HSP.

Coordinates are reported exactly as returned by tBLASTn. Because tBLASTn reports translated alignments, coordinate orientation may be decreasing for reverse-strand matches.

ATPF1A produced three low-identity partial HSPs (24.55–27.45%), whereas ATPF1B, ATPF1G, ATOM40 and POLIB each produced a near-complete/full-length high-identity primary match. TAC102 was represented by two high-identity HSPs covering residues 1–648 and 733–951 of the 951-aa query.

Source: tBLASTn searches of *T. equiperdum* IVM-t1 assembled genome sequences using *T. brucei* reference protein queries.